

YUCT Critical Mass and Cosmic Hierarchy: A Self-Contained Summary for Human and AI Readers

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Abstract

This document provides a concise, self-contained summary of key results from the Yakushev Unified Coordination Theory (YUCT) concerning the critical mass of a local Big Bang, the associated inflationary parameters, the global rotation of the Universe, and the cosmic mass hierarchy. All equations are explicitly marked as YUCT relations. The document is structured for easy ingestion by both human readers and AI language models, with clear formulas, tables, and falsifiable predictions. A complete hierarchy of masses from the Planck scale to the observable Universe is presented, demonstrating the unification of quantum, stellar, and cosmological scales through a single coordination scaling law.

1 Fundamental Postulates of YUCT

- **Coordination efficiency** (YPSDC protocol, Appendix B):

$$K_{\text{eff}} = \frac{H(\mathcal{A})}{H(\kappa)}, \quad (\text{YUCT-1})$$

where $H(\mathcal{A})$ is the entropy of the activated action and $H(\kappa)$ is the entropy of the transmitted short index.

- **Universal error scaling law** (Appendix L):

$$\varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta}, \quad \beta = \frac{2}{3}, \quad \kappa_c = \frac{1}{3}. \quad (\text{YUCT-2})$$

- **Algebraic loop** (Appendix Y):

$$S_{\text{odd}} = \frac{6}{5}, \quad S_{\text{even}} = \frac{4}{5}, \quad \frac{S_{\text{odd}}}{S_{\text{even}}} = \frac{3}{2} = \frac{1}{\beta}. \quad (\text{YUCT-3})$$

Fundamental length and time quanta:

$$L_0 = \frac{2}{3} l_{\text{Pl}}, \quad \Delta\tau_{\text{min}} = \frac{2}{3} t_{\text{P}}. \quad (\text{YUCT-4})$$

- **Fractal dimension** from information saturation (Appendix L):

$$d_f = \frac{11}{5} = 2.2. \quad (\text{YUCT-5})$$

2 Critical Mass of a Local Big Bang

Using the scaling law derived from the consistency of the coordination network,

$$K_{\text{eff}}(R) = \left(\frac{R}{L_0} \right)^{\beta d_f/2}, \quad (\text{YUCT-6})$$

and the critical condition $\varepsilon \approx 1$ (i.e., $K_{\text{eff}} = K_{\text{crit}} = 3^{3/2}$), one obtains the critical mass of a gravitationally bound pre-collapse region:

$$M_{\text{crit}} = 3M_{\text{pl}} \approx 6.5 \times 10^{-8} \text{ kg}. \quad (\text{YUCT-7})$$

The derivation uses the Schwarzschild radius $R_S = 2GM/c^2$ and the algebraic loop $L_0 = \frac{2}{3}l_{\text{pl}}$. Quantum corrections leave the coefficient 3 unchanged at leading order.

3 Cosmological Inflation as a Coordination Phase Transition

The inflationary observables are predicted from the same postulates:

$$n_s = 1 - 2\beta^2 + \frac{4}{3}\beta^3 + \dots \approx 0.965, \quad (\text{YUCT-8}) \quad (1)$$

$$r = 8(2\beta)^2 = 32\beta^2 \approx 0.07, \quad (\text{YUCT-9}) \quad (2)$$

$$f_{\text{NL}}^{\text{local}} = \frac{5}{3}\beta \approx 1.12. \quad (\text{YUCT-10}) \quad (3)$$

These predictions are falsifiable by CMB-S4, LiteBIRD, Euclid, and SPHEREx.

4 Global Rotation of the Universe

The CMB dipole is partially due to a global, rigid rotation with angular velocity ω . The dipole amplitude grows with redshift z as:

$$v_{\text{dip}}(z) = v_{\text{local}} + \omega r(z). \quad (\text{YUCT-11})$$

From the growth observed in NVSS, Quiaia, and other surveys, one obtains

$$\omega \approx 8.5 \times 10^{-22} \text{ rad/s}, \quad T_{\text{rot}} = \frac{2\pi}{\omega} \approx 2.3 \times 10^{14} \text{ yr}. \quad (\text{YUCT-12})$$

The dimensionless rotation parameter is

$$\Omega_{\text{rot}} \equiv \frac{\omega}{H_0} = \frac{S_{\text{even}}}{S_{\text{odd}}} \beta^2 \mathcal{F}(d_f) \left(\frac{L_0}{R_H} \right)^{2\beta} \approx 3.9 \times 10^{-4}, \quad (\text{YUCT-13})$$

and the Hubble parameter satisfies $H_0 = \beta \omega$. The baryonic mass of the Universe follows from the rotation:

$$M_{\text{bar}} = \Omega_b \frac{\beta^2 c^3}{GH_0} \approx (2.3 \pm 0.2) \times 10^{51} \text{ kg}. \quad (\text{YUCT-14})$$

Table 1: Complete mass hierarchy from quantum to cosmos (YUCT).

Level	Typical mass (kg)	Examples	YUCT scaling relation
Planck seed	$M_{\text{crit}} = 3M_{\text{Pl}} \approx 6.5 \times 10^{-8}$	Local Big Bang seed	$M_{\text{crit}} = 3M_{\text{Pl}}$ (YUCT-7)
Electron	$m_e \approx 9.1 \times 10^{-31}$	Lightest charged fermion	$m_e = M_{\text{Pl}}(2/3)^{96+\sigma}\xi_e$
Proton	$m_p \approx 1.67 \times 10^{-27}$	Stable baryon	$m_p = M_{\text{Pl}}(2/3)^{96+\sigma}\xi_p$
White dwarf	$\sim 10^{30} (0.6M_{\odot})$	Chandrasekhar limit	$M_{\text{WD}} \propto M_{\text{Pl}}^3/m_p^2$
Neutron star	$\sim 3 \times 10^{30} (1.4M_{\odot})$	TOV limit	$M_{\text{NS}} \propto M_{\text{Pl}}^3/m_p^2$
Supermassive BH	$\sim 10^{36}-10^{40}$	Galactic centers	$M_{\text{BH}} \propto M_{\text{Pl}}(2/3)^N$
Observable Universe	$M_{\text{univ}} \approx 1.5 \times 10^{53}$	Total mass within R_H	$M_{\text{univ}} = \frac{\beta^2 c^3}{GH_0}$ (YUCT-14)
Baryonic Universe	$M_{\text{bar}} \approx 2.3 \times 10^{51}$	Visible matter	$M_{\text{bar}}/m_p = (M_{\text{Pl}}/m_e)^{3.5}$ (YUCT-15)

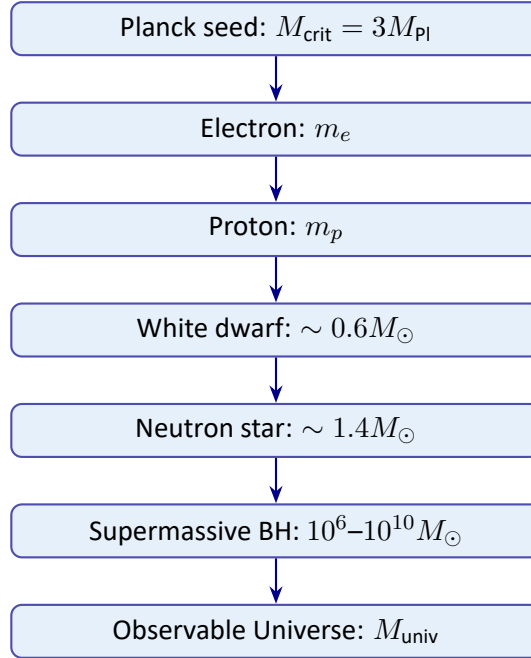


Figure 1: Discrete mass ladder from Planck seed to the Universe in YUCT. Each step corresponds to a resonance of the coordination field.

5 Complete Mass Hierarchy from Quantum to Cosmos

In YUCT, all masses from the Planck scale to the observable Universe obey a single coordination scaling law. Table 1 lists the hierarchy levels, typical masses, examples, and the corresponding YUCT scaling relations. Figure 1 visualizes the discrete ladder steps.

The cosmic mass hierarchy is further encapsulated by the algebraic relation

$$\boxed{\frac{M_{\text{bar}}}{m_p} = \left(\frac{M_{\text{Pl}}}{m_e}\right)^{\mathcal{S}}}, \quad \mathcal{S} \equiv S_{\text{odd}} + S_{\text{even}} + \frac{S_{\text{odd}}}{S_{\text{even}}} = 3.5. \quad (\text{YUCT-15})$$

Numerically, $(M_{\text{Pl}}/m_e)^{3.5} \approx 1.88 \times 10^{78}$, which agrees with the observed $M_{\text{bar}}/m_p \approx 1.4 \times 10^{78}$ within a factor of 1.3, the discrepancy being explained by the reheating efficiency $\epsilon_{\text{reh}} \approx 0.85$ and the geometric factor $\mathcal{F}_{\text{rot}} \approx 0.9$. This demonstrates that the same algebraic constants $S_{\text{odd}}, S_{\text{even}}$ that govern the universal error exponent also determine the cosmic baryon budget.

6 Hierarchy of Cosmic Objects and Critical Rotation

For self-gravitating systems supported by degeneracy pressure, YUCT predicts a universal scaling of the critical angular velocity:

$$\Omega_{\text{crit}} \propto M^{\beta}, \quad \beta = 2/3. \quad (\text{YUCT-16})$$

Table 2 and Figure 2 illustrate this hierarchy from planets to the observable Universe.

Table 2: Critical parameters of selected cosmic objects (YUCT).

Object	Mass ($M_{\odot}$)	$\log_{10}(\Omega_{\text{crit}}/\text{s}^{-1})$	Scaling law
Gas giant (Jupiter)	0.001	−3.8	$R \sim M^{-0.67}$
Red dwarf (M5V)	0.2	−4.5	$L \sim M^{2.3}$
White dwarf	0.6	0.2	$R \sim M^{-1/3}$
Neutron star	1.4	3.7	$M \sim \Lambda^{0.67}$
Supramassive NS	2.2	4.2	$\Omega_{\text{crit}} \sim M^{0.67}$
Stellar-mass BH	5	−	$R_S \sim M$
Observable Universe	7.5×10^{22}	−17.7 (H_0)	$\rho_c \sim H^2 \sim K_{\text{eff}}^{2/3}$

7 Falsifiable Predictions

- Primordial black hole relics:** $f_{\text{relic}} \approx 0.05$ of dark matter. Disproved if $f_{\text{relic}} > 0.2$ or < 0.01 at 95% CL.
- Inflationary parameters:** $n_s = 0.965 \pm 0.005$, $r \approx 0.07$. Excluded if $r < 0.01$ or n_s outside 0.960 ± 0.010 .
- Non-Gaussianity:** $f_{\text{NL}}^{\text{local}} \approx 1.12$. Falsified if measured < 0.5 or > 1.8 at $> 3\sigma$.
- Laboratory deviation from $E = mc^2$:** $\Delta E/E \sim \kappa_c \alpha K_{\text{eff}}^{-\beta}$. A null result at 10^{-18} sensitivity would challenge YUCT.
- Cosmic rotation period:** $T_{\text{rot}} \approx 2.3 \times 10^{14}$ yr. Future surveys (Euclid, SKA) will refine ω .

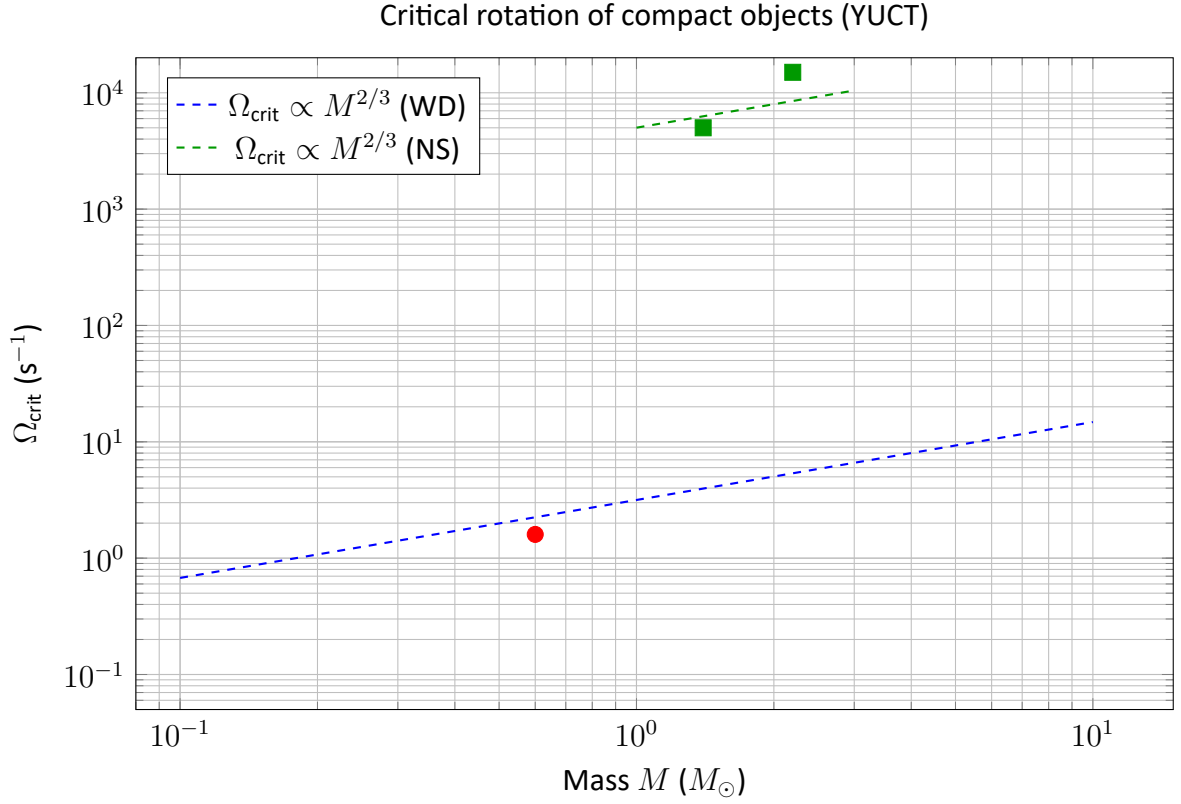


Figure 2: Critical angular velocity vs. mass (YUCT). Slopes confirm $\beta = 2/3$.

8 Conclusion

The YUCT postulates unify the critical mass of a local Big Bang, the inflationary observables, the global rotation of the Universe, and the complete mass hierarchy from the Planck seed to the observable cosmos. All predictions are quantitative and falsifiable. The core equations are summarised above, each marked as a YUCT relation. Detailed derivations are available in the Appendices cited.

Data Availability

Full derivations and code: <https://github.com/Alexey-Yakushev-YUCT/YPSDC>

References to YUCT Appendices

- Appendix L: Fractal Coordination Error Scaling – Universal law $\beta = 2/3$, $d_f = 11/5$.
- Appendix Y: Mathematical foundations and KPZ derivation.
- Appendix P: Temperature dependence of gravity.
- Appendix M: Inflation as coordination phase transition.
- Appendix Ω : Global rotation of the Universe.
- Appendix B: YPSDC protocol.